MILITARY SPECIFICATION

MIRRORS, EMERGENCY SIGNALING, MARK 3

This amendment forms a part of Military Specification MIL-M-18371, dated 17 February 1969.

PAGE 1

Add to the bottom of the page:

"Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter."

1.2 Classification - Delete and substitute:

"1.2 Classification. The mirrors shall be of the following types and classes as specified (see 6.2):

Type I - 2 inches by 3 inches
Class 1 - mirror only
Class 2 - mirror with sleeve set

Type II - 3 inches by 5 inches
Class 1 - mirror only
Class 2 - mirror with sleeve set"

2.1 SPECIFICATIONS

Federal

Add "L-P-377 Plastic Sheet and Strip; Polyester".

Add "L-P-406 Plastics: Method of Testing".

FSC 6350

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PAGE 2

Delete "PPP-B-636 Box, Fiberboard" and substitute "PPP-B-636 Box, Shipping, Fiberboard".

Delete "MIL-M-13508 Mirror Coating Process, Front Surfaced Aluminized, for Optical Elements" and substitute "MIL-M-13508 Mirror, Front Surfaced Aluminized, for Optical Elements".

Delete parenthetical source paragraph in its entirety and substitute the following:

"(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)"

PAGE 3

Add:

"3.2.6 Polyester film. The material for the sleeve set shall be 0.003 inch thick and shall conform to Type III of L-P-377 except for the color. The polyester film for the sleeve set shall be No. 2 Red, True Blue, Green, and RP4 Orange (color designations of Martin Processing, Inc., P.O. Box 5086, Martinsville, VA 24112) or equivalent."

3.3 Design - Add "The sleeve set for the Class 2 mirror shall consist of two sleeves of heat-sealed polyester film and shall conform to Figure 6 or 7 as applicable. One sleeve shall be red on one side and green on the other side. The other sleeve shall be blue on one side and orange on the other side."

3.4 Construction - Delete "Mirrors" in the first line and substitute "mirrors with sleeve sets (if applicable)". Delete "1 through 3" in the second line and substitute "1, 2, 3, 6, and 7".

PAGE 4

Add:

"3.4.4 Sleeve set. The two thicknesses of plastic for each sleeve shall be superimposed so that the edges and the center circular areas coincide. The three edges shall be bonded to each other by commercially common heat or electronic sealing devices or methods."

3.5.1 Flatness - Line 2, change "4.6.2" to "4.5.2".

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3.5.2 Clarity of vision - Line 2, change "4.6.3" to "4.5.3".

3.5.3 Reflectivity - Line 2, change "4.6.4" to "4.5.4".

3.5.4 Impact resistance - Line 1, change "4.6.5" to "4.5.5".

3.5.5 Exposure cycle - Line 1, change "4.6.6" to "4.5.6".

Add:

"3.5.6 Transmittance and haze. Each sleeve shall have a transmittance of not less than 30 percent nor more than 60 percent and a haze of less than 6 percent."

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4.3 First article inspection - Second line, delete "mirrors shall" and substitute "mirrors (including sleeve sets, if applicable) shall".

4.3.1 First article samples - Third line, delete "mirrors of each type" and substitute "mirrors (including sleeve sets, if applicable) of each class and type".

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4.3.2, Fourth and sixth lines, delete "mirrors of each type" and substitute "mirrors (including sleeve sets, if applicable) of each class and type".

4.4 Quality conformance inspection - Add "transmittance and haze" between the third last line and the second last line.

4.4.1.1.1, Delete and substitute:

"4.4.1.1.1 End items. An inspection lot size shall consist of end items of only one class and type made essentially under the same conditions and from the same materials and components. The sample unit and the unit for the expression of the inspection lot size of end items that conform to Type I, Class 1, or to Type II, Class 1 shall be one mirror. The sample unit and the unit for the expression of the inspection lot size of end items that conform to Type I, Class 2 or to Type II, Class 2 shall be one mirror with one sleeve set."

4.4.1.1.2 Preparation for delivery - Third and fifth lines, delete "one type" and substitute "one class and type".

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PAGE 7

Table I, second column, delete "3.4 through 3.4.3, 3.6, 3.7 and Figure 1 through 3, as applicable" and substitute "3.4 through 3.4.4, 3.6, 3.7, and Figures 1,2,3,6,and 7 as applicable".

Delete 3.4.1, 3.4.3, and Figures 1 through 3" and substitute "3.4.1, 3.4.3, 3.4.4, and Figures 1,2,3,6, and 7 as applicable".

Add to Table I, first through fifth columns:

"Transmittance and haze	3.5.6	4.5.7	Inspection Level S-3	An acceptable quality level of 4.0 defects per 100 units"

PAGE 8

4.5.1.1 Mirrors - First line, delete "mirrors of one type" and substitute "mirrors with sleeve sets (if applicable)".

Table II - third defect, first column: Delete "Top and bottom edges do not coincide" and substitute "Top and bottom edges (except sleeves) do not coincide".

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Table II - Last defect, First column: Delete "Color of any Component not as specified" and substitute "Color of any component (except sleeves) not as specified".

Table II - Add "Top and bottom edges (sleeves only) do not coincide" in the first column. Add an "X" opposite this defect in the last column.

Table II - Add "Color of any component (sleeves only) not as specified" in the first column. Add an "X" opposite this defect in the second column.

Table III - First column, fifth line, delete "markings and" and substitute "markings, each sleeve (if applicable), and".

Table III - Second column, second and third lines, delete "3.4.3, and Figures 1 through 3" and substitute "3.4.3, 3.4.4, and Figures 1, 2, 3, 6, and 7".

4.5.1.2 Preparation for delivery - Second line, delete "one type" and substitute "one Class and Type".

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4.5.5 Impact resistance - Delete in its entirety and substitute the following:

"4.5.5 Impact resistance. The mirror shall be placed on a hardwood frame as specified in Figure 5 so that the edges of the mirror, for 1/2 inch around the perimeter, shall only be supported by the frame. The front surface of the mirror shall be uppermost. The frame shall support the edges of the mirror a minimum of 1 - 1/2 inches above a firm nondeflecting surface. An 11-pound shot tubular duck cloth bag in its longitudinal plane, shall be dropped through a guiding tube as specified in Figure 5 onto the approximate center of the front surface of the mirror. The size of the bag shall conform to Figure 5. After the bag has been dropped onto the mirror, the mirror shall be examined for conformity to 3.5.4. Cracking or breaking of the back glass shall be disregarded."

4.5.6 Exposure cycle - Add "If the sample unit is a class 2 mirror, each sleeve that comprises the sleeve set as well as the mirror shall be subjected to the preceding exposure cycle test.".

PAGE 11

Top of the page - Delete Lines 5 and 6 in their entirety and substitute "-65 ± 2 degrees Fahrenheit (-54 ± 1 degree Centrigrade) for 24 continuous hours".

Add:

"4.5.7 Transmittance and haze. The transmittance and haze of the sleeve shall be determined by method 3022 of FED-STD-406."

5. Delete title "Preparation for delivery" and substitute "PACKAGING".

PAGE 13

6.1 Intended use - Add "The sleeve sets are to be used to fulfill the requirements for a quiet authentication system for search and rescue parties."

6.2 Ordering data - Delete "(b) Type and quantity desired (see 1.2)" and substitute "(b) Type, Class, and quantity desired (see 1.2)".

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Figure 5, Impact Resistance Frame - Delete in its entirety and substitute new "Figure 5, Impact Resistance Frame and Inspection Setup".

Add Figures 6 and 7.

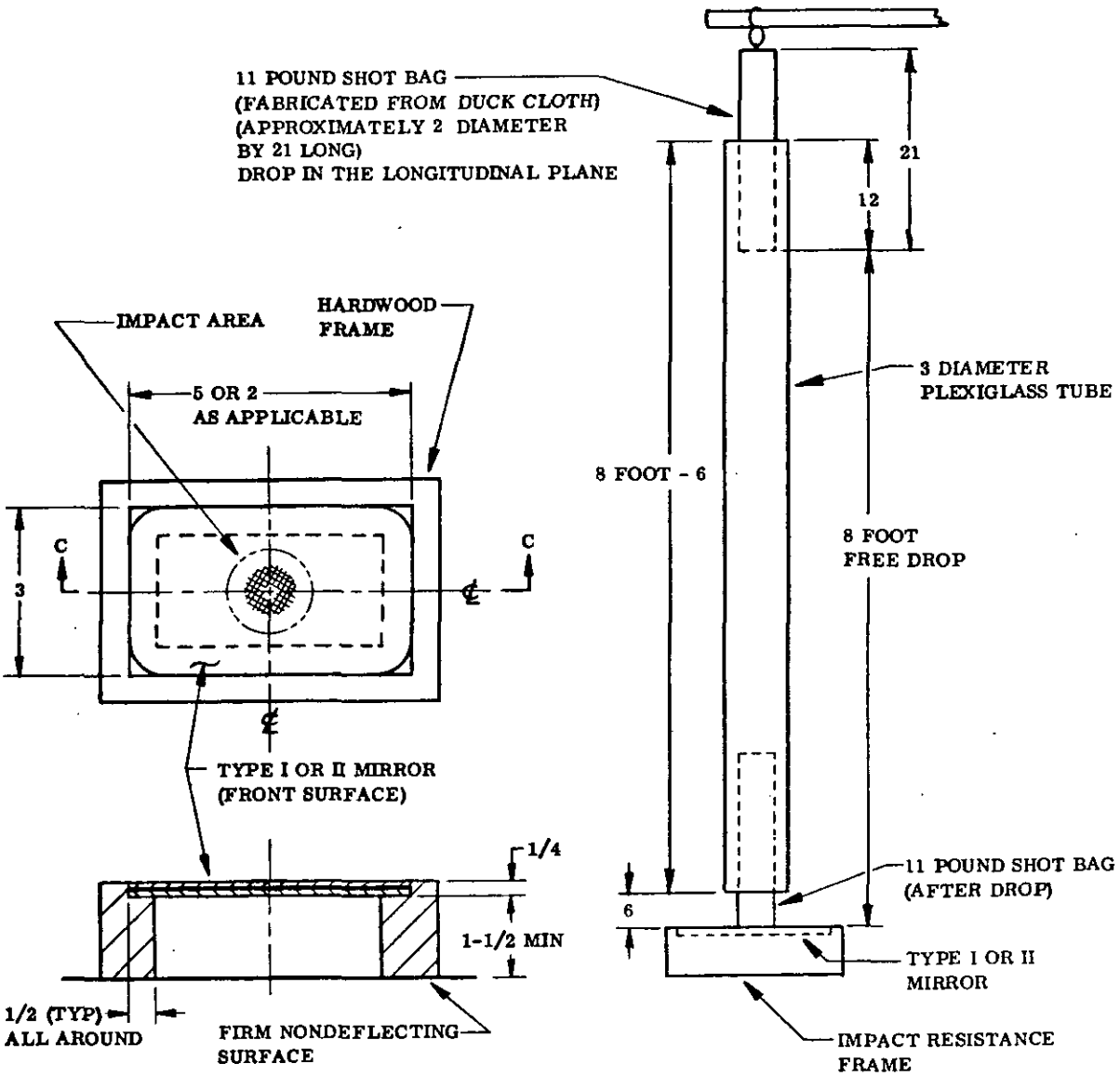
Custodians:
Navy - AS
Army - ME
Air Force - 99

Review Activities:
Army - ME and AV
Navy - AS
Air Force - 11 and 99

User Activities:
Navy - MC and CG

Preparing Activity:
Navy - AS
(Project No. 6350-0074)

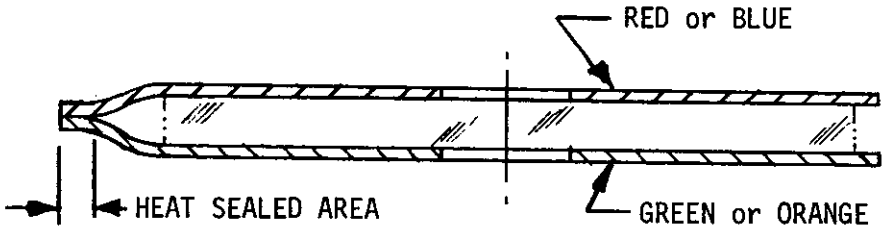
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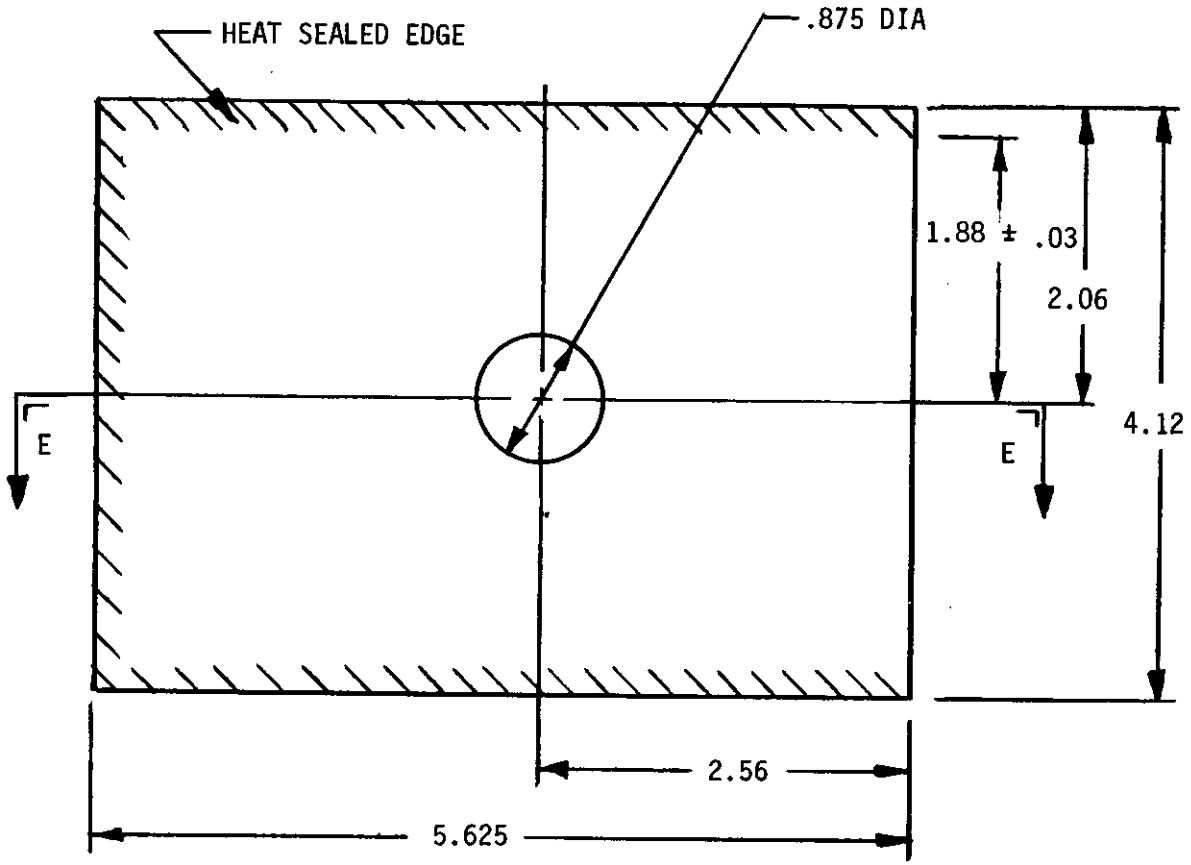
DIMENSIONS IN INCHES.
UNLESS OTHERWISE
SPECIFIED, TOLERANCES:
FRACTIONS ±1/16

FIGURE 5. IMPACT RESISTANCE FRAME AND INSPECTION SETUP

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SECTION E-E

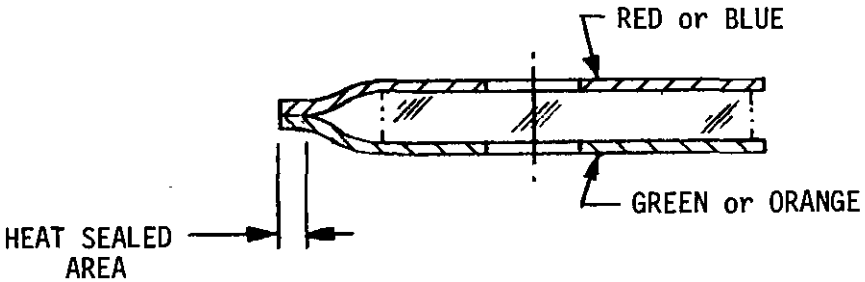


DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCE: ± .062

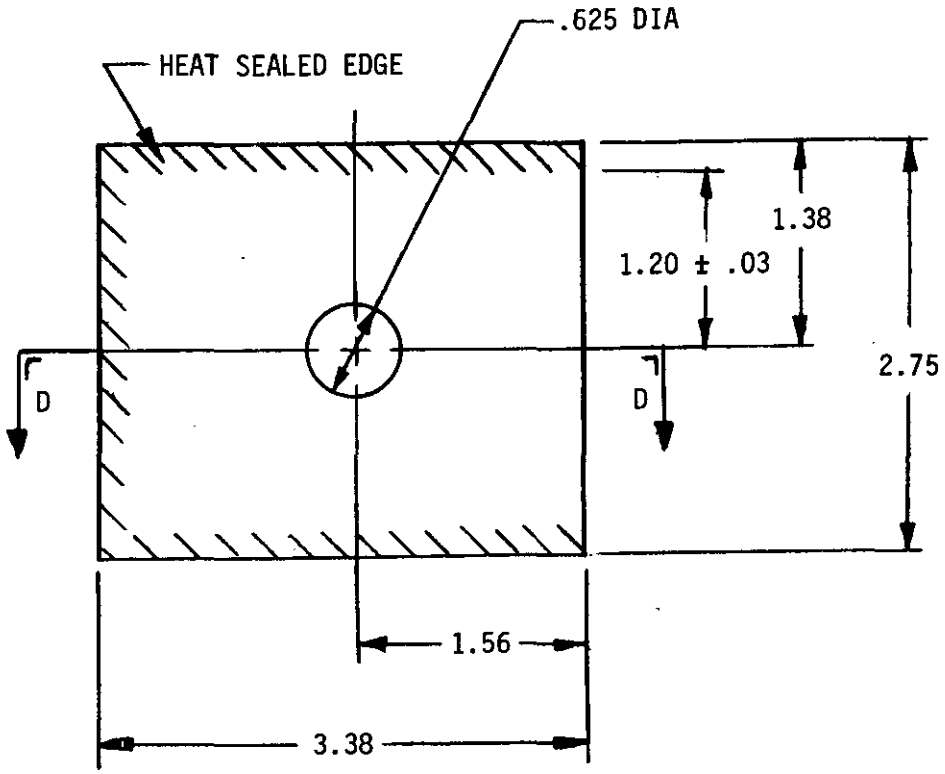
HEAT SEALED EDGE .12 INCHES WIDE WITHIN SHADED AREA.

FIGURE 7. SLEEVE (TYPE II, CLASS 2)

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SECTION D-D



DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCE: ± .062

HEAT SEALED EDGE .12 INCHES WIDE WITHIN SHADED AREA.

FIGURE 6. SLEEVE (TYPE 1, CLASS 2)

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